

DBMS UNIT 3

UNIT – III Question Bank Categorized by Topic

1. Overview of the SQL Query Language

Q1 (06 Marks)

Discuss the basic clauses required to execute a SQL Query with syntax and example.

Q2 (06 Marks)

Explain the 6 clauses of an SQL retrieval query. Discuss which of the clauses are required and which are optional.

Q3 (10 Marks)

Describe the six clauses in the syntax of an SQL retrieval query. Show what type of constructs can be specified in each of the six clauses. Which of the six clauses are required and which are optional?

2. SQL Data Definition (DDL)

Q4 (10 Marks)

Consider the following schema:

SAILORS(SID, SNAME, RATING, AGE)

BOATS(BID, BNAME, BCOLOR)

RESERVES(SID, BID, DAY)

Formulate the SQL queries for the following.

i. Create the above tables by imposing the following constraints

Rating can be an integer between 1 and 10

- ii. Find the names of sailors who have reserved Yellow boat but not Orange boat
- iii. Find the names of sailors who have not reserved Interlake boats
- iv. For each red boat find the number of reservations made.

Q5 (08 Marks)

Consider the following schema.

SAILORS(SID,SNAME,RATING,AGE)

BOATS(BID,BNAME,BCOLOR)

RESERVES(SID,BID,DAY)

Formulate the SQL queries for the following.

1. Create the above tables by imposing the following constraints
 - i. Rating can be an integer between 1 and 10 only
 - ii. Boats name should be 'Jackpot', 'Zameen', 'Interlake'
2. Find the names of sailors who have reserved Yellow boat but not Orange boat
3. Find the names of sailors who have not reserved Interlake boats
4. Find the names of sailors who are older than oldest sailor with a rating of 10
5. For each red boat find the number of reservations made.

Q6 (10 Marks)

Explain the purpose and syntax of the following commands in SQL by considering a relational schema that has at least one numerical attribute.

- i) Delete
- ii) Update
- iii) Alter
- iv) INSERT

Q7 (10 Marks)

Explain the purpose and syntax of the following commands in SQL by considering a relational schema that has at least one numerical attribute.

- i) Delete
- ii) Update
- iii) Alter
- iv) Grant
- v) Revoke

Q8 (06 Marks)

Write SQL syntax for the following with example:

- i) SELECT
 - ii) ALTER
 - iii) UPDATE
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3. Basic Structure of SQL Queries

Q9 (10 Marks)

Consider the schema for employee database below.

Employee (employee-name, street, city)

Works (employee-name, company-name, salary)

Company (company-name, city)

Manages (employee-name, manager-name)

Give an expression in SQL for each of the following queries.

- i. Find the company that has the most employees.
- ii. Find the company that has the smallest payroll.
- iii. Define a view consisting of manager-name and the average salary of all employees who work for that manager.

iv. Write CHECK condition to ensure that every employee works for a company located in the same city as the city in which the employee lives.

Q10 (10 Marks)

Consider the following tables:

BRANCH(branch-name:string, branch-city:string, assets:real)

ACCOUNT(accno:int, branch-name:string, balance:real)

DEPOSITOR(customer-name:string, accno:int)

CUSTOMER(customer-name:string, customer-street:string, customer-city:string)

LOAN(loannumber:int, branch-name:string, amount:real)

BORROWER(customer-name:string, loan-number:int)

Write the SQL for the following:

- i. List in alphabetical order all customers who have a loan at 'Perryridge' branch.
 - ii. Find the names of all customers who street address includes the substring "Main".
 - iii. List the loan numbers in descending order of amount.
 - iv. Find the average account balance at each branch.
 - v. Find the name of the customer who has the highest balance in the bank.
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Q11 (08 Marks)

Consider the following tables:

BRANCH(branch-name, branch-city, assets)

Customer(customer-name, customer-street, customer-city)

loan(loan-number, branch-name, amount)

Borrower(customer-name, loan-number)

Account(account-number, branch-name, balance)

Depositor(customer-name, account-number)

Write the SQL Queries for the following:

- i. List in alphabetical order all customers who have a loan at 'MSRIT' branch.
 - ii. Find the names of all customers who area address includes the substring "yesvanthpur".
 - iii. List the loan numbers in ascending order of amount.
 - iv. Find the average loan amount at each branch.
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Q12 (08 Marks)

Consider the following tables:

BRANCH (branch-name:string, branch-city:string, assets:real)

ACCOUNT (accno:int, branch-name:string, balance:real)

DEPOSITOR (customer-name:string, accno:int)

CUSTOMER(customer-name:string, customer-street:string, customer-city:string)

LOAN (loannumber:int, branch-name:string, amount:real)

BORROWER (customer-name:string, loan-number:int)

Write the SQL Queries for the following:

- i. List in alphabetical order all customers who have a loan at 'Perryridge' branch.
 - ii. Find the names of all customers who street address includes the substring "Main".
 - iii. List the loan numbers in descending order of amount.
 - iv. Find the average account balance at each branch.
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Q13 (06 Marks)

Consider the following relation schema

Works(Pname,Cname,salary)

Lives(Pname,Street,City)

located_in (Cname, city)

Manager(Pname,Mgrname)

Write the SQL queries for the following

- i) Find the names of all persons who live in the city Bangalore.
 - ii) Retrieve the names of all person of "Infosys" whose salary is between Rs .50000
 - iii) List the names of the people who work for "Tech M" along with the cities they live in.
 - iv) Find the average salary of "Infosys" persons.
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Q14 (04 Marks)

Consider the following relations for a database that keeps track of business trips of salespersons in a sales office.

SALESPERSON(SID,Name, Start_year, Dept_no)

TRIP(SID, From_city, To_city, Departure_date, Return_date, Trip_id)

EXPENSE(Trip_id, Account#, Amount)

Write SQL Queries for the following

- i. List the Sid and the name of the salesman who made trips to 'San Francisco'.
 - ii. List the total trip expenses incurred by the salesman with name "RIO".
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4. Additional Basic Operations

Q15 (06 Marks)

Explain any three clauses of SELECT FROM WHERE in SQL with an example.

Q16 (06 Marks)

Explain any three different clauses of SELECT in SQL with an example.

Q17 (06 Marks)

Explain different set operations in SQL with an example for each.

Q18 (06 Marks)

Write note on impedance mismatch.

Q19 (05 Marks)

Specify three different ways of inserting value into the table. Create a table in SQL which fulfil the following criteria's:

- i) Specifying attribute constraints
 - ii) Specifying key and referential constraints.
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5. Null Values

Q20 (08 Marks)

Illustrate the following with syntax and examples:

- i. On delete Set NULL and On Update Cascade
- ii. Group by and Having

Q21 (08 Marks)

Illustrate the following with syntax and examples:

- i. On delete Set NULL and On Update Cascade
 - ii. IS NULL.
 - iii. Group by and having.
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6. Aggregate Functions

Q22 (06 Marks)

Explain the use of group-by and having clause in SQL with an example.

Q23 (05 Marks)

Explain GROUP BY function in SQL with an example.

Q24 (09 Marks)

Consider the following table with the details of products sold in a shop for two days.

PRODUCTS

Date	ItemName	Unit_Sold
9/3/23	Pencils	100
9/3/23	Pen	500
10/3/23	Pencils	50
10/3/23	Pen	900
10/3/23	Eraser	75

Construct SQL queries for the following scenarios:

- Display the total number of items sold for each item name.
 - Display the item names and total number of items sold for only those items where total of sold items is greater than 1000.
 - Display the table sorted in the descending order on the number of quantities sold.
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7. Nested Sub Queries

Q25 (06 Marks)

What do you mean by nested queries in SQL? Why IN operator is preferred over JOIN in nested queries? Explain with an example.

Q26 (04 Marks)

Differentiate nested and co-related nested queries with an example for each.

Q27 (06 Marks)

Discuss the correlated nested queries with an example.

8. Modification of the Database

Q28 (08 Marks)

Consider the schema given below:

EMP(Ssn, fname, lname, bdate, address, sex, salary, superssn, dno)

DEPT(dnumber, dname, mgrssn, mgrstartdate)

PROJECT(pnumber, pname, ploc, dnum)

WORKS_ON(essn, pno, hours)

Write the SQL statements for the following queries:

- i. Select the names of all female employees whose first letter of first name is 'J' and fourth letter is 'H'.
 - ii. For each project on which more than two employees work, retrieve the project number, the project name, and the number of employees who work on that project.
 - iii. Retrieve the names of all employees whose supervisor's supervisor has '888445555' as his Ssn.
 - iv. Update the Dumber of 'Research' Department from 10 to 15. Explain any constraint violations occur. If so, how to resolve that?
 - v. Show how to add a new column named "middlename" be added to EMP table.
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Q29 (08 Marks)

Consider the schema given below:

EMP(Ssn, fname, lname, bdate, address, sex, salary, superssn, dno)

DEPT(dnumber, dname, mgrssn, mgrstartdate)

PROJECT(pnumber, pname, ploc, dnum)

WORKS_ON(essn, pno, hours)

DEPENDENT(Dname, Essn)

Mention SQL query for the following:

- i. To change the location and controlling department number of project number 10 to "Bellaire" & 5 respectively.
- ii. Retrieve all employees in department 5 whose salary is between \$30,000 & \$40,000.

- iii. Retrieve a list of employees & the projects they are working on, ordered by department & within each department ordered alphabetically by last name, first name.
 - iv. Retrieve name of employees with no dependents.
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Q30 (06 Marks)

Consider the Supply-Parts database given below. Assume relevant data types for attributes.

SUPPLIER (Sid, Sname, Address)

PART (PID, Pname, Color)

SHIPMENT (Sid, PID, Cost)

Mention SQL query for the following

- i. Find the Sid's of suppliers who supply a yellow part.
 - ii. For every supplier print the name of the supplier and the total number of parts that he/she supplies.
 - iii. Update the part color supplied by supplier s3 to green.
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Q31 (06 Marks)

Consider the schema given below:

EMP(Ssn, fname, lname, bdate, address, sex, salary, superssn, dno)

DEPT(dnumber, dname, mgrssn, mgrstartdate)

PROJECT(pnumber, pname, ploc, dnum)

WORKS_ON(essn, pno, hours)

DEPENDENT(Dname, Essn)

Write the SQL statements for the following queries:

- i. List the names of all department managers who have no dependents
 - ii. Increase the salary of employees of research department by 5%
 - iii. Delete employees of mechanical department.
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9. Join Expressions

Q32 (06 Marks)

Differentiate between JOIN ON and NATURAL JOIN operations in SQL.

Q33 (06 Marks)

Differentiate between JOIN and NATURAL JOIN operations in SQL.

Q34 (08 Marks)

Explain with examples, the various JOIN operations possible in SQL.

10. Views

Q35 (06 Marks)

Illustrate how are views created and deleted in SQL. Give examples.

Q36 (05 Marks)

Explain specification, implementation and update of views in SQL with example.

Q37 (06 Marks)

Differentiate between two approaches of view implementation.

Q38 (10 Marks)

Consider the schema for employee database below.

Employee (employee-name, street, city)

Works (employee-name, company-name, salary)

Company (company-name, city)

Manages (employee-name, manager-name)

Give an expression in SQL for each of the following queries.

iii. Define a view consisting of manager-name and the average salary of all employees who work for that manager.

11. Integrity Constraints

Q39 (10 Marks)

How does SQL allow implementation of the entity integrity and referential integrity constraints? Give one example for each.

Q40 (06 Marks)

Explain the need for assertions. Create an assertion in SQL for the following statement and explain.

"The salary of an employee must not be greater than the salary of the manager of the department that the employee works for"

Q41 (06 Marks)

What is Trigger? What is its need? How to create assertion for a relation? Explain with examples.

Q42 (10 Marks)

Consider the schema for employee database below.

Employee (employee-name, street, city)

Works (employee-name, company-name, salary)

Company (company-name, city)

Manages (employee-name, manager-name)

Give an expression in SQL for each of the following queries.

iv. Write CHECK condition to ensure that every employee works for a company located in the same city as the city in which the employee lives.

12. SQL Data Types and Schemas

Q43 (06 Marks)

List the data types that are allowed for SQL attributes.

13. Comprehensive SQL Query Programs

Q44 (12 Marks)

STUDENT (regno: string, name: string, major: string, bdate:date)

COURSE (course#:int, cname: string, dept: string)

ENROLL (regno: string, course#:int, sem:int, marks:int)

BOOK_ADOPTION (course#: int, sem: int, book-ISBN:int)

TEXT (book-ISBN:int, book-title: string, publisher: string, author: string)

Write SQL queries for the following:

- i) List the students name who have enrolled for 2nd sem.
 - ii) List no of books adopted in semester '4'.
 - iii) List no of books adopted by CSE department.
 - iv) List the maximum no of courier adopted by any student.
 - v) List the courses where course name begins with 'a' being held by 'cs' department.
 - vi) List department which have more than 2 courses.
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Q45 (10 Marks)

Employee (Name, SSN, address, Sex, salary, DNo)

Department (Dname, Dnumber, MGRSSN, MGRSTARTDATE)

Dept_locations (Dnumber, Dlocations)

Project (Pname, pnumber, plocation, Dnum)

Work_on (ESSN, PNO, Hours)

Dependent (ESSN, Dependent_name, sex, B_date, Relationship)

Write the queries in SQL to

- i. Retrieve the name and address of all the employees who work for the “research” department.
 - ii. Find the names of the employee who works on all projects controlled by dnumber 5.
 - iii. List all the projects on which the employee “smith” is working.
 - iv. Retrieve the names of the employee who have no dependents.
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Q46 (10 Marks)

Consider the following schema of Company database. Write the SQL queries for the following.

EMP(Name,SSN,Salary,SuperSSN,Gender,Dno)

DEPT(DNum,Dname,MgrSSN,Dno)

DEPT_LOC(Dnum,Dlocation)

DEPENDENT(ESSN,Dep_name,Sex)

WORKS_ON(ESSN,Pno,Hours)

PROJECT(Pname,Pnumber,Plocation,Dnum)

- i) Retrieve the name of each employee who works on all the projects controlled by department number 5 (use non-Exists)
 - ii) Retrieve the Social Security numbers of all employees who work on project numbers 1, 2, or 3
 - iii) Retrieves the name and address of every employee who works for the ‘Research’ department
 - iv) Find the sum of the salaries of all employees of the ‘Research’ department, as well as the maximum salary, the minimum salary, and the average salary in this department.
 - v) For each department, retrieve the department number, the number of employees in the department, and their average salary.
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Q47 (08 Marks)

Consider the schema given below for a aircraft database.

Aircraft (aircraft ID, aircraft name, cruising range)

Certified (emp ID, aircraft ID)

Employee (Emp ID, ename, salary)

Write the SQL statements for the following queries:

- i. Find the employee ID's of employee who make the highest salary.
 - ii. Find the employees who are certified for the largest no of aircrafts.
 - iii. Arrange the flight no with respect to the ascending order of distance.
 - iv. Find the name of pilots who can operate flights with a range greater than 3000 miles but are not certified on any Boeing aircraft.
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Q48 (10 Marks)

Consider the following relations containing airline flight information:

Flights(flno: integer, from: string, to: string, distance: integer, departs: time, arrives: time)

Aircraft(aid: integer, aname: string, cruisingrange: integer)

Certified(eid: integer, aid: integer)

Employees(eid: integer, ename: string, salary: integer)

Note that the Employees relation describes pilots and other kinds of employees as well; every pilot is certified for some aircraft (otherwise, he or she would not qualify as a pilot), and only pilots are certified to fly.

Write the following queries using SQL:

- i. Find the names of aircraft such that all pilots certified to operate them earn more than 80,000.
 - ii. Print the enames of pilots who can operate planes with cruisingrange greater than 3,000miles.
 - iii. Find the eids of pilots certified for Boeing aircraft.
 - iv. Find the names of pilots certified for Boeing aircraft.
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Q49 (10 Marks)

What is an embedded SQL? For the relational schema given below, write C function to retrieve the names of pilots certified for a given aircraft using embedded SQL statements

Flights(flno: integer, from: string, to: string, distance: integer, departs: time, arrives: time)

Aircraft(aid: integer, aname: string, cruisingrange: integer)

Certified(eid: integer, aid: integer)

Employees(eid: integer, ename: string, salary: integer)

Note that the Employees relation describes pilots and other kinds of employees as well; every pilot is certified for some aircraft (otherwise, he or she would not qualify as a pilot), and only pilots are certified to fly.